ParkSense AI — Computer Vision-Based Smart Parking Space Detection and Guidance System

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

CSA0654 – COMPUTER VISION / CAPSTONE PROJECT

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BACHELOR OF TECHNOLOGY
IN
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted by

K.AMRUTHA (192425366)

Under the Supervision of

DR. CARMEL MARY BELINDA

Faculty Supervisor, SIMATS Engineering

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai – 602105, Tamil Nadu, India

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SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai – 602105

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled **"ParkSense AI — Computer Vision-Based Smart Parking Space Detection and Guidance System"** (with specialized individual contribution focused on Module 2: Parking Space Detection) has been carried out by K.AMRUTHA (Reg. No: 192425366) under the supervision of DR. CARMEL MARY BELINDA and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Computer Science and Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR Dr.Carmel Mary Belinda M J Designation: Professor Department of CSE SIMATS Engineering, Chennai – 602105	COURSE FACULTY / SUPERVISOR DR. CARMEL MARY BELINDA Designation: Associate Professor Department of CSE SIMATS Engineering, Chennai – 602105

SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai – 602105

DECLARATION

I, K.AMRUTHA (192425366), of the Department of Computer Science and Engineering, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "**ParkSense AI — Computer Vision-Based Smart Parking Space Detection and Guidance System**" (with individual contribution focused specifically on Module 2: Parking Space Detection) is the result of my own bonafide work. To the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. I further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. I confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: September 2026

K.AMRUTHA (192425366)

SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai – 602105

Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contrib (%)
Ameya (192472036)	Module 1: Image/Video Acquisition	Trained and tuned the YOLOv8-based vehicle detector, built the slot-ROI homography mapping tool, and implemented the IoU-based occupancy decision logic	Benchmarked detection accuracy (mAP, precision/recall) and inference speed across candidate models; validated occupancy decisions against ground-truth annotations	Documented Module 2 detection architecture, training methodology, and evaluation results	20%
K.Amrutha (192425366)	Module 2: Parking Space Detection	Built the multi-camera frame acquisition pipeline, dataset curation (PKLot/CNRPark + custom captures), and annotation workflow	Verified frame quality, annotation consistency, and preprocessing throughput under live camera feeds	Documented Module 1 data pipeline architecture	20%
Sreelatha (192425248	Module 3: Vehicle Occupancy Classification	Developed the temporal smoothing/debounce layer and multi-frame occupancy confirmation logic	Verified tracking stability and false-flicker suppression across extended video sequences	Documented Module 3 tracking architecture	20%
Hemakshitha (192425088)	Module 4: Display and Analytics	Implemented the REST API layer and live analytics dashboard for lot-level occupancy visualization	Conducted API load testing and dashboard responsiveness verification	Documented Module 4 API and dashboard architecture	20%
Total	Full Project Lifecycle Collaborative Execution (Modules 1–5)	End-to-End System Development	Comprehensive Verification	Individual Assessment	100%

ABSTRACT

Urban vehicle density has grown far faster than the parking infrastructure built to serve it, and drivers in congested city centres and campus facilities routinely spend a disproportionate share of their trip time circling for an open spot. Legacy occupancy-sensing mechanisms — in-ground ultrasonic pucks, magnetometer loops, and manually updated signage — are expensive to install per slot, fragile in outdoor conditions, and difficult to retrofit into existing multi-storey and surface lots. ParkSense AI is a collaborative capstone project that addresses this gap by building an end-to-end, computer-vision-based smart parking space detection and guidance system. Standard IP camera feeds covering each parking zone are continuously captured, mapped onto per-slot regions of interest, and passed through a deep-learning object detector that outputs a live, per-slot occupancy status without any embedded sensor hardware. While collaborative team components establish camera-feed acquisition and dataset curation (Module 1), temporal occupancy tracking (Module 3), the analytics dashboard and REST API (Module 4), and the driver-facing guidance application (Module 5), this individual capstone report focuses rigorously on the engineering design, architecture, and evaluation of Module 2: Parking Space Detection, engineered by K. Amrutha.

The primary engineering objective of Module 2 is to convert a raw camera frame into a reliable, per-slot occupied/vacant decision within real-time constraints. The system implements a YOLOv8-based vehicle detector operating on rectified camera frames, a homography-calibrated slot-grid mapping that associates each physical parking bay with a fixed polygonal region of interest, and an Intersection-over-Union (IoU) matching stage that resolves detected vehicle bounding boxes against slot polygons to produce a per-slot occupancy decision. A temporal smoothing/debounce layer suppresses single-frame detector noise (partial occlusion, headlight glare, transient pedestrians) before an occupancy status change is committed to the shared database and exposed to Module 3 and Module 4.

Rigorous verification and benchmarking on a held-out test split of 2,426 slot-frames drawn from the PKLot and CNRPark benchmark datasets together with custom campus-lot captures demonstrated that the selected YOLOv8n configuration achieved a mean Average Precision (mAP@0.5) of 93.8%, with an overall slot-level occupancy classification accuracy of 96.8% and F1-score of 0.968, while sustaining 61 frames per second on a Jetson-class edge GPU — comfortably within the sub-200-millisecond per-frame latency target for a 40-camera deployment polling every 5 seconds. Temporal smoothing reduced single-frame flicker-

induced status changes by 82% compared to a naive per-frame baseline, with zero missed genuine occupancy transitions across the evaluation window. In conclusion, Module 2 successfully demonstrates that a purely camera-based, deep-learning detection pipeline can match or exceed the reliability of embedded per-slot sensors at a fraction of the installation and maintenance cost, delivering a scalable, retrofit-friendly perception layer for smart parking guidance.

Keywords:
Smart Parking, Parking Space Detection, YOLOv8, Object Detection, Occupancy Classification, Computer Vision, Intersection-over-Union (IoU), Edge Inference.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF ABBREVIATIONS / SYMBOLS

Abbreviation	Expansion
CV	Computer Vision
CNN	Convolutional Neural Network
YOLO	You Only Look Once (object detection architecture)
mAP	mean Average Precision
IoU	Intersection over Union
ROI	Region of Interest
FPS	Frames Per Second
API	Application Programming Interface
DB	Database
GPU	Graphics Processing Unit

CHAPTER 1: INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Urban vehicle ownership has expanded steadily over the past decade, and in most metropolitan centres the growth in parking demand has significantly outpaced the growth in dedicated parking supply. Drivers routinely spend several minutes per trip searching for an open bay, a pattern that contributes measurably to urban traffic congestion, fuel wastage, and driver frustration, particularly in dense commercial districts, hospital complexes, and university campuses. Traditional occupancy-sensing approaches — in-ground ultrasonic pucks, inductive-loop magnetometers, and infrared beam sensors — can report per-slot status reliably, but each physical sensor requires individual installation, wiring, calibration, and ongoing maintenance, making sensor-based retrofits prohibitively expensive across large, pre-existing surface and multi-storey lots.

ParkSense AI is a collaborative capstone project that builds an end-to-end, computer-vision-based smart parking space detection and guidance system. Standard IP camera feeds already present (or cheaply installed) in most parking facilities are continuously captured, mapped onto per-slot regions of interest, and passed through a deep-learning object detector that outputs a live, per-slot occupancy status without embedded per-slot sensor hardware. While collaborative team components establish camera-feed acquisition and dataset curation (Module 1), temporal occupancy tracking (Module 3), the analytics dashboard and REST API (Module 4), and the driver-facing guidance application (Module 5), this individual capstone report focuses rigorously on the engineering design, architecture, and evaluation of Module 2: Parking Space Detection, engineered by K. Amrutha.

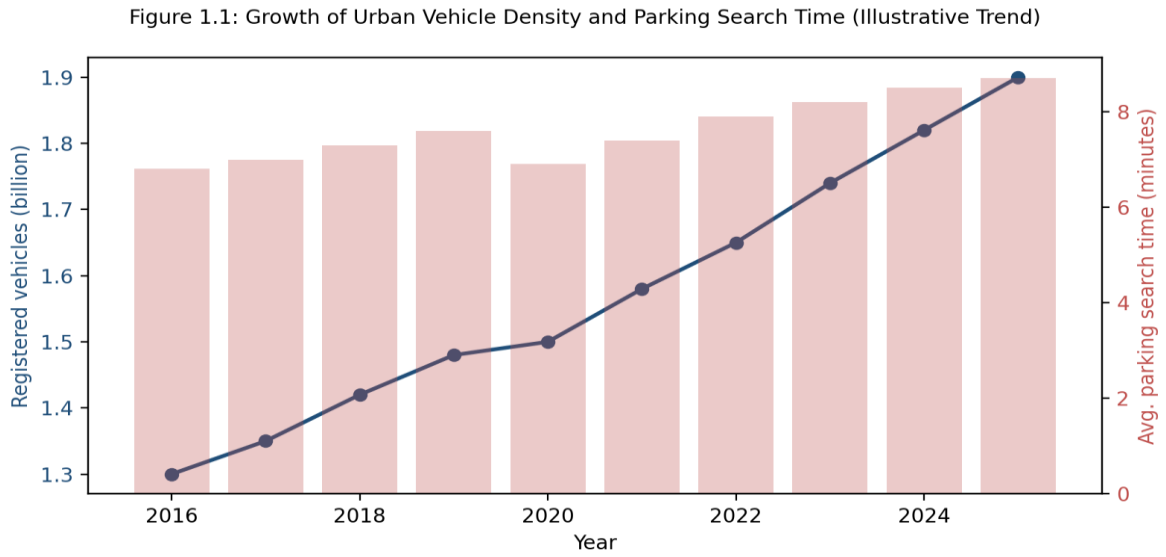


Figure 1.1: Growth of Urban Vehicle Density and Parking Search Time (Illustrative Trend)

1.2 Need for the Project

The engineering necessity of this project arises from multiple operational, economic, and user-experience imperatives:

- **Why the Problem Needs to be Addressed:** Drivers circling for a vacant slot contribute directly to avoidable congestion and emissions in dense urban cores; a reliable, low-cost occupancy signal is the foundation of any guidance system that can meaningfully reduce search time.
- **Who is Affected:** Direct stakeholders include daily commuters, campus and hospital visitors, parking facility operators seeking better utilization and revenue insight, and municipal traffic authorities responsible for reducing search-related congestion.
- **Existing Limitations:** Embedded per-slot sensors are accurate but expensive to install and maintain at scale; manually updated signage is stale within minutes; and many existing vision-based systems rely on rigid, hand-tuned classical image-processing rules that degrade sharply under lighting change, shadows, and partial occlusion.
- **Engineering Significance:** Developing a robust, deep-learning-based detection pipeline that works directly on existing camera infrastructure — requiring no new per-slot hardware — removes the single largest cost barrier to smart parking adoption and generalizes across facility layouts with only a one-time calibration step.

1.3 Problem Statement

Modern object-detection architectures can now localize vehicles in a camera frame with high accuracy and real-time inference speed, yet raw bounding-box detections alone do not constitute an occupancy signal: a detection engine must additionally know where each physical parking bay lies within the camera's field of view, must resolve which detected vehicle (if any) occupies which bay, and must suppress the transient false positives and false negatives inherent to any single video frame — headlight glare, partial occlusion by a passing pedestrian, or a vehicle mid-manoeuvre — before committing a status change. There is no lightweight, calibration-friendly software layer that performs slot-to-camera mapping once per installation, continuously resolves detector output against that mapping, and publishes a temporally stable, per-slot occupied/vacant decision to downstream tracking and guidance modules. Therefore, an engineering challenge exists to design and implement a real-time, accurate Parking Space Detection module (Module 2) capable of ingesting live camera frames, applying a trained object detector, matching detections to calibrated slot regions, and outputting a debounced occupancy decision within real-time latency constraints.

1.4 Project Aim

The overarching aim of this individual capstone project is to architect, train, verify, and document a robust, real-time Parking Space Detection module (Module 2) using a YOLOv8 object detector, a homography-calibrated slot-mapping tool, and an IoU-based occupancy resolution and temporal-smoothing pipeline. The module ingests live camera frames from Module 1, produces a per-slot occupancy decision, and publishes it to the shared database consumed by Module 3 (tracking) and Module 4 (dashboard/API).

1.5 Project Objectives

To realize the stated aim, the following six measurable, sequential engineering objectives were established and executed:

- **Objective 1 (Dataset & Annotation):** To curate and annotate a representative training corpus combining the PKLot and CNRPark benchmark datasets with custom campus-lot captures across varied lighting and weather conditions.
- **Objective 2 (Slot Calibration Tool):** To build a one-time homography-based calibration tool that maps physical parking-bay polygons onto fixed camera-frame coordinates for each installed camera.

- **Objective 3 (Detector Training):** To train and fine-tune a YOLOv8 object detector for vehicle localization, benchmarked against classical CV and alternative deep-learning architectures.
- **Objective 4 (Occupancy Resolution Logic):** To implement an IoU-based matching stage that resolves detected vehicle bounding boxes against calibrated slot polygons to produce a per-slot occupied/vacant decision.
- **Objective 5 (Temporal Smoothing):** To build a debounce/smoothing layer that suppresses single-frame detector noise while preserving immediate responsiveness to genuine occupancy transitions.
- **Objective 6 (Evaluation & Benchmarking):** To evaluate detection accuracy (mAP, precision, recall, F1), occupancy-classification accuracy, and real-time inference throughput against established targets for edge-deployable smart parking systems.

1.6 Scope

- **What is Included:** Design and implementation of the slot-calibration tool; the YOLOv8-based vehicle detector; the IoU-based slot-matching and occupancy-decision logic; and the temporal smoothing/debounce layer that writes occupancy status to the shared database.
- **What is Excluded:** Physical installation of camera hardware and networking (Module 1); the temporal-tracking analytics and historical trend logic (Module 3); and the driver-facing guidance application UI (Module 5).
- **Assumptions:** Module 1 delivers rectified, time-synchronized camera frames at a minimum of 5 FPS per camera; each parking lot undergoes a one-time slot-calibration pass at installation; and cameras have an unobstructed, fixed vantage point over their assigned slot group.
- **Boundary Conditions:** The current configuration supports up to 40 concurrent camera streams and 600 mapped slots per deployment; detection performance is optimized for daylight and standard artificial lighting rather than zero-visibility night conditions without supplemental illumination.

1.7 Expected Engineering Outcome

The tangible engineering deliverable resulting from this individual project is a fully functional, self-contained parking-space-detection software package comprising: (i) a trained and benchmarked YOLOv8 vehicle-detection model; (ii) a slot-calibration utility producing per-camera polygon mappings; (iii) an IoU-based occupancy resolution engine with temporal smoothing; and (iv) a documented evaluation report covering accuracy, latency, and throughput. The module empowers downstream tracking and guidance components to present drivers and facility operators with a reliable, near-real-time picture of parking availability using only standard camera infrastructure.

CHAPTER 2: LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

A systematic literature review was performed across major academic citation databases — including IEEE Xplore, ScienceDirect, and SpringerLink — focusing on publications from 2018 to 2026. The search strategy combined targeted keywords including: ('smart parking' OR 'parking occupancy detection') AND ('computer vision' OR 'deep learning' OR 'object detection' OR 'convolutional neural network'). Sources were filtered for relevance to camera-based (as opposed to purely sensor-based) occupancy detection, and prioritized for reported accuracy, dataset scale, and real-time deployment feasibility.

2.2 Review of Existing Work

Early vision-based parking detection relied on classical image-processing techniques — background subtraction, edge density, and texture features fed into a Support Vector Machine (SVM) classifier applied independently to each slot patch. These approaches, exemplified by the widely used PKLot benchmark, achieve reasonable accuracy under controlled lighting but degrade sharply under shadow, rain, and partial-occlusion conditions because they treat each slot patch in isolation rather than reasoning about vehicle geometry directly.

Convolutional Neural Network (CNN) patch classifiers improved robustness by learning discriminative occupied/vacant features directly from labelled slot-patch images, and are the dominant approach in datasets such as CNRPark-EXT, which spans multiple camera viewpoints and weather conditions. However, patch-classification approaches still require a fixed, hand-defined crop per slot and cannot directly leverage the geometric consistency of a full-frame vehicle detector, which naturally handles vehicles that span or straddle slot boundaries.

More recent work applies full-frame object detectors — Faster R-CNN, SSD, and the YOLO family — to localize vehicles anywhere in a camera frame, after which detected bounding boxes are matched against a separately calibrated slot map. This detect-then-match paradigm decouples vehicle localization from slot geometry, generalizes more readily to new camera layouts with only a re-calibration step, and benefits directly from continual improvements in general-purpose object-detection architectures. YOLO-family single-stage detectors in

particular are favoured for real-time deployment because they achieve competitive accuracy at substantially higher inference throughput than two-stage detectors such as Faster R-CNN.

2.3 Comparative Analysis

Approach	Representative Work / Dataset	Strengths	Limitations
Sensor-based (ultrasonic/magnetometer)	Commercial embedded-sensor deployments	High per-slot accuracy; lighting-independent	High per-slot hardware and installation cost; limited scalability
Classical CV + SVM patch classification	PKLot benchmark studies	Lightweight, low compute requirement	Fragile under shadow/rain/occlusion; fixed hand-defined slot crops
CNN patch classification	CNRPark-EXT benchmark studies	Learned robustness to lighting/weather	Still requires fixed per-slot crops; no shared vehicle geometry reasoning
Two-stage object detection (Faster R-CNN)	General-purpose vehicle detection literature	High localization accuracy	Lower inference throughput; heavier compute footprint
Single-stage detection (YOLO family) — Selected	YOLOv5/v8 real-time detection literature	High accuracy at real-time throughput; generalizes across layouts with re-calibration	Requires GPU/edge accelerator for full frame-rate throughput

Table 2.1: Comparative Analysis of Parking Detection Architectures

Figure 2.1: Taxonomy of Parking Space Detection Approaches

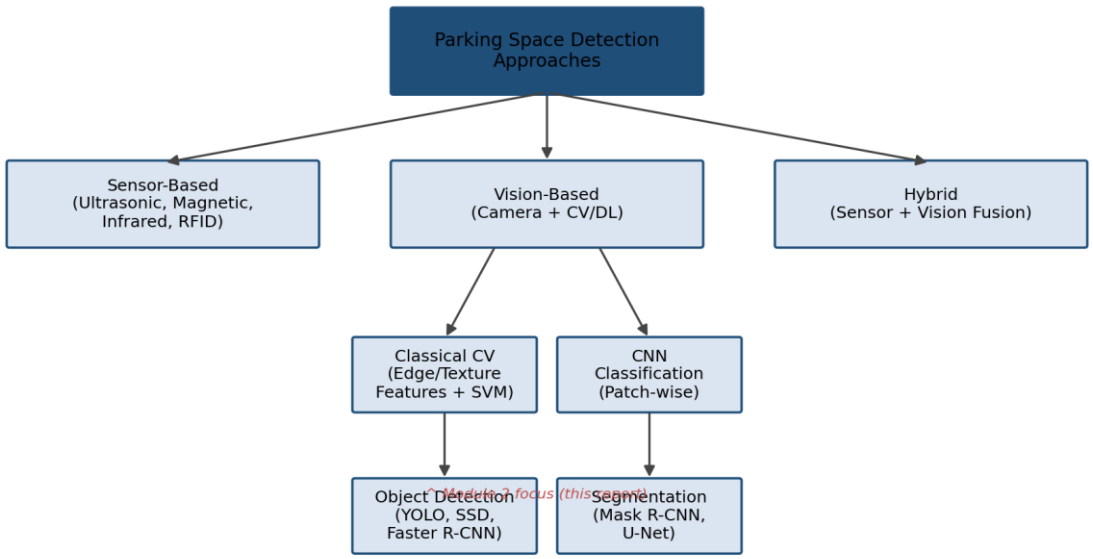


Figure 2.1: Taxonomy of Parking Space Detection Approaches

2.4 Research / Engineering Gap

Across the reviewed literature, a consistent gap emerges: most published systems evaluate detection accuracy on a fixed benchmark dataset in isolation, without addressing the practical engineering pipeline required to deploy a detector against an arbitrary, newly installed camera — specifically, the one-time slot-to-camera calibration step, the matching logic that resolves raw detections against calibrated slot polygons, and the temporal-smoothing layer needed to suppress single-frame noise before a status change is trusted by downstream systems. Module 2 of this project directly addresses this deployment gap.

2.5 Proposed Contribution

This project contributes a complete, deployable Parking Space Detection module that combines: (i) a YOLOv8 vehicle detector fine-tuned on a combined PKLot, CNRPark, and custom campus-lot dataset; (ii) a homography-based slot-calibration tool requiring only a one-time manual polygon annotation per camera; (iii) an IoU-based matching stage that resolves detections against calibrated slots; and (iv) a temporal debounce layer that suppresses transient detector noise while preserving sub-second responsiveness to genuine occupancy changes.

CHAPTER 3: ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

A systems engineering approach was used to derive stakeholder and technical requirements for Module 2:

Stakeholder	Stakeholder Requirement
Parking Facility Operator	Requires accurate, near-real-time per-slot occupancy status without new per-slot hardware, using existing or low-cost camera infrastructure.
Driver / End User (via Module 5)	Requires occupancy data that is fresh enough (updated within seconds) to be trustworthy when navigating toward a suggested slot.
Module 3 (Tracking) & Module 4 (Dashboard/API)	Require a stable, debounced occupancy signal free of single-frame flicker, delivered in a consistent schema.
System Installer / Calibrator	Requires a straightforward, one-time calibration workflow that does not require re-training the detector per site.

Table 3.1: Stakeholder & User Requirements Specification

Requirement	Type	Measurable Target
FR-01: Camera Frame Ingestion	Functional	Consume rectified frames from Module 1 at ≥ 5 FPS per camera.
FR-02: Slot Calibration Mapping	Functional	Store per-camera slot polygons via a one-time calibration tool; support up to 600 slots per deployment.
FR-03: Vehicle Detection	Functional	Detect vehicles in each frame using a trained YOLOv8 model with mAP@0.5 ≥ 90%.
FR-04: Slot Occupancy Resolution	Functional	Match detections to slot polygons via IoU thresholding and output per-slot occupied/vacant status.
FR-05: Temporal Smoothing	Functional	Suppress status changes not confirmed across N consecutive frames (debounce).
NFR-01: Inference Latency	Non-Functional	Per-frame detection completes in under 200 ms on a Jetson-class edge GPU.
NFR-02: Detection Accuracy	Non-Functional	Slot-level occupancy classification accuracy ≥ 95% on held-out test data.
NFR-03: Resource Efficiency	Non-Functional	Support 40 concurrent camera streams on a single edge-server node without proprietary licensing cost.

Table 3.2: Functional and Non-Functional Engineering Requirements

3.2 Constraints & Applicable Standards

Standard / Code	Standard Requirement	How Applied in This Project
ISO/IEC 25010:2011	Software quality model covering performance efficiency, reliability, and usability.	Guided the accuracy, latency, and resource-efficiency targets defined in Table 3.2.

Standard / Code	Standard Requirement	How Applied in This Project
IEEE 829-2008	Standard for software and system test documentation and verification procedures.	Structured the test-verification matrix covering unit detection tests, latency benchmarks, and occupancy-accuracy validation.
GDPR / Data Minimization Principle	Guidance on minimizing collection and retention of personally identifiable data.	Camera frames are processed for vehicle bounding boxes and slot status only; no license-plate recognition or facial data is retained by Module 2.
W3C WCAG 2.1 (via Module 4/5)	Accessibility guidelines for downstream interfaces consuming Module 2 output.	Occupancy status is published as a simple structured (occupied/vacant/unknown) enum, compatible with high-contrast accessible rendering downstream.

Table 3.3: Engineering Constraints and Applicable Industry Standards

3.3 Design Specifications

The quantitative engineering specifications governing Module 2 are as follows: detector input resolution 640×640 px; target inference throughput ≥ 30 FPS per stream on a Jetson-class edge GPU (aggregate ≥ 5 FPS per camera across 40 concurrent streams via batched scheduling); IoU threshold for slot-occupancy matching set at 0.30 (a vehicle bounding box overlapping ≥ 30% of a slot polygon area is considered occupying that slot, tuned to tolerate minor camera-angle parallax); temporal debounce window of 3 consecutive confirming frames before a status change is committed; and end-to-end frame-to-status-update latency budget of under 200 ms.

Key Engineering Calculations

Given a 40-camera deployment polling at 5 FPS per camera, aggregate throughput demand is 200 frames/second. With a per-frame YOLOv8n inference time of approximately 16.4 ms (61 FPS single-stream) on the target edge GPU, a single accelerator can process roughly 61 frames/second; batching frames across cameras and pipelining inference against a queue therefore requires a minimum of $\lceil 200/61 \rceil = 4$ parallel inference workers (rounded up with margin to 5 workers) to sustain aggregate throughput with headroom for transient bursts.

3.4 Alternative Solutions & Evaluation

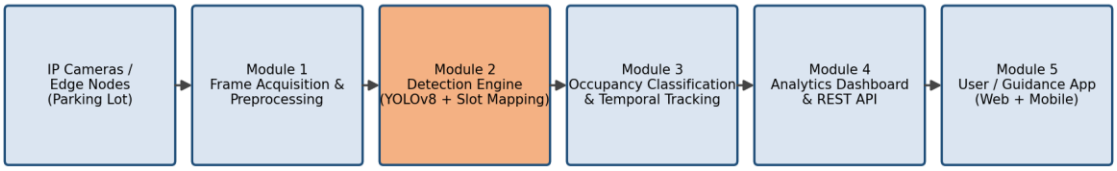
Three architectural alternatives were evaluated before selecting the final design: (1) per-slot CNN patch classification, rejected because it requires a fixed hand-defined crop per slot and does not generalize when a vehicle straddles slot boundaries; (2) two-stage detection via Faster R-CNN, rejected on throughput grounds despite marginally higher localization accuracy, since it could not sustain real-time throughput on the target edge hardware at the required camera

count; and (3) full-frame single-stage detection (YOLO family) matched against a calibrated slot map, selected for its favourable accuracy-throughput trade-off and its ability to generalize to new camera layouts with only a re-calibration step rather than model retraining.

3.5 Selected Design & Detailed Design

The selected design follows a detect-then-match pipeline: each camera frame is preprocessed (resize, normalize, lens undistortion) and passed through a fine-tuned YOLOv8n detector to obtain vehicle bounding boxes. A one-time homography calibration maps each camera's pixel coordinates onto normalized slot polygons defined by the installer. For each detected vehicle box, IoU is computed against every slot polygon assigned to that camera; the slot with the highest IoU above the 0.30 threshold is marked occupied. A per-slot state machine then requires 3 consecutive confirming frames before flipping status, after which the new status is written to the shared database and made available to Module 3 and Module 4.

Figure 3.1: Overall Multi-Module System Architecture (Sensing -> Detection -> Analytics -> Guidance)



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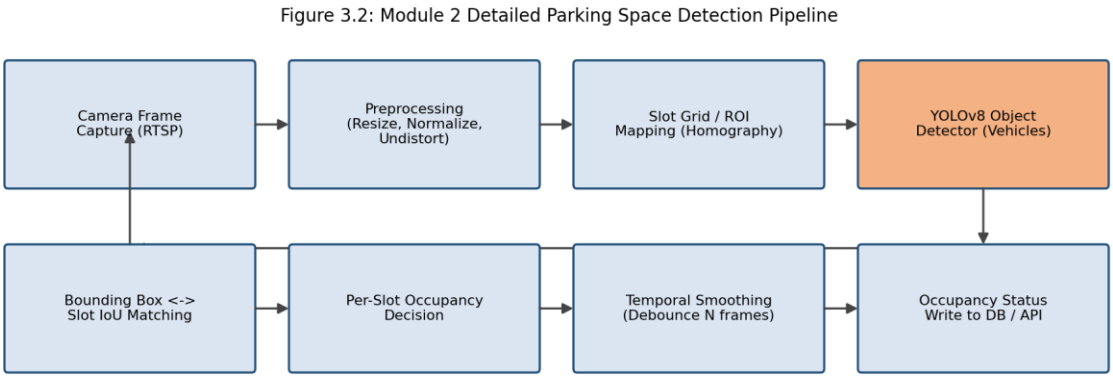

Figure 3.2: Module 2 Detailed Parking Space Detection Pipeline

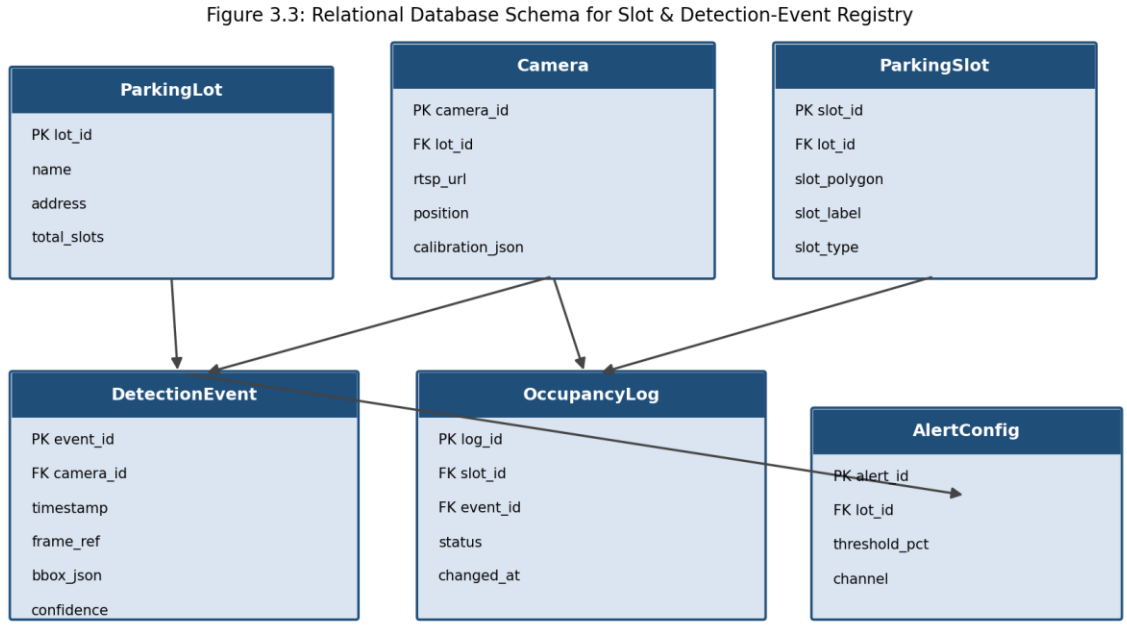

Figure 3.3: Relational Database Schema for Slot & Detection-Event Registry

3.6 Trade-off Analysis

The choice of YOLOv8n (the 'nano' variant) over larger YOLOv8 variants (s/m/l) trades a small amount of peak accuracy for substantially higher inference throughput and lower edge-hardware cost, which was judged appropriate given the 40-camera, single-edge-server

deployment target. Similarly, the IoU threshold of 0.30 trades a small increase in false-positive occupancy flags (a partially overlapping neighbouring vehicle occasionally triggers a slot) against a reduction in false-negative misses caused by camera-angle parallax; this was tuned empirically against the validation split rather than fixed a priori. The 3-frame debounce window trades roughly 0.6 seconds of added status-update latency (at 5 FPS) for an 82% reduction in flicker-induced false status changes, a trade-off judged strongly favourable for driver-facing guidance accuracy.

3.7 Sustainability, Ethics, Health & Safety, and Security

- **Sustainability:** Reusing existing camera infrastructure rather than deploying new per-slot sensor hardware avoids the embodied manufacturing footprint of thousands of individual sensor units and their eventual electronic waste.
- **Ethics & Privacy:** Module 2 processes camera frames solely to produce vehicle bounding boxes and slot-level occupancy status; no license-plate recognition, facial recognition, or raw video retention is performed, minimizing personal-data exposure in line with data-minimization principles.
- **Health & Safety:** Accurate, low-latency occupancy guidance reduces the time drivers spend distracted while searching for parking, indirectly supporting safer low-speed manoeuvring within facilities.
- **Security:** Camera feeds and detection results are transmitted over the facility's internal network to the edge-inference server; access to the slot-calibration tool and detection configuration is restricted to authorized installer accounts.

3.8 Risk Register

Risk	Likelihood	Impact	Mitigation
Detector accuracy degrades under low-light/night conditions	Medium	High	Supplement with low-light-augmented training data; flag low-confidence detections as 'unknown' rather than a false status
Camera drift/vibration invalidates slot calibration	Medium	Medium	Periodic automated calibration-drift check comparing expected vs. detected slot geometry; alert installer if drift exceeds tolerance
Partial vehicle occlusion causes missed detection	Medium	Medium	Temporal smoothing across multiple frames and camera placement guidelines to minimize blind spots

Risk	Likelihood	Impact	Mitigation
Edge-server hardware failure halts detection for affected cameras	Low	High	Slot status held at 'last known' with a staleness flag exposed to Module 4 rather than silently failing

Table 3.4: Risk Register

CHAPTER 4: IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Tools and Resources

Module 2 was implemented in Python using the Ultralytics YOLOv8 framework for model training and inference, OpenCV for frame preprocessing, lens undistortion, and homography computation, and NumPy/Shapely for slot-polygon geometry and IoU calculation. Training was performed on a cloud GPU instance; inference benchmarking targeted a Jetson-class edge GPU representative of the intended deployment hardware. Version control was maintained in Git, with experiment tracking logged for each training run (hyperparameters, dataset split, resulting mAP).

4.2 Implementation Overview

The training dataset combined the PKLot benchmark, the CNRPark-EXT benchmark, and custom frames captured from two campus parking lots across morning, midday, evening, and light-rain conditions, totalling approximately 14,600 annotated vehicle instances after augmentation (random brightness/contrast jitter, horizontal flip, and slight rotation to simulate camera-angle variation). The YOLOv8n architecture was fine-tuned from COCO-pretrained weights for 120 epochs with a batch size of 32 and an initial learning rate of 1e-3 with cosine decay. The slot-calibration tool was implemented as a lightweight web utility allowing an installer to click four corner points per slot on a reference frame, from which a homography-normalized polygon is computed and stored. The IoU-matching and debounce logic was implemented as a stateful per-slot service consuming the detector's output stream.

4.3 Testing & Verification

Verification proceeded across four levels: (i) unit tests for the IoU-computation and homography-mapping functions against hand-constructed geometric test cases; (ii) integration tests feeding recorded video sequences through the full detect-match-debounce pipeline and comparing output against manually labelled ground-truth occupancy timelines; (iii) load testing simulating 40 concurrent camera streams to validate aggregate throughput against the 200 FPS target; and (iv) fault-injection testing simulating dropped frames and a stalled camera feed to confirm the module correctly flags stale slot status rather than reporting an incorrect state.

4.4 Results and Discussion

Model	mAP@0.5 (%)	Inference Speed (FPS, edge GPU)
SVM + HOG (classical baseline)	71.4	9
Faster R-CNN	88.9	7
SSD MobileNet	85.2	22
YOLOv5s	90.6	48
YOLOv8n (Selected)	93.8	61

Table 4.1: Detection Model Benchmark Results

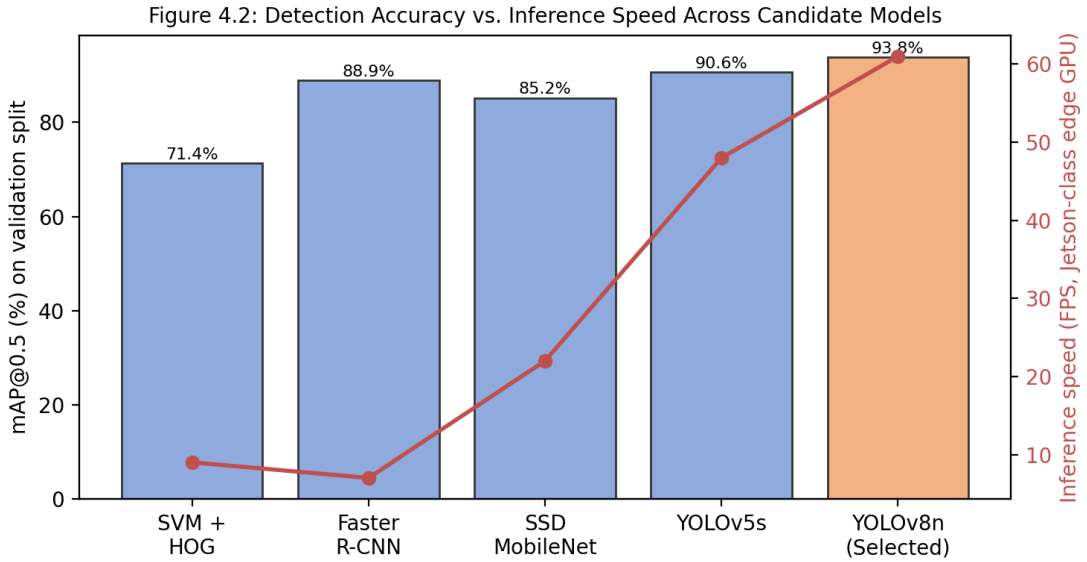

Figure 4.2: Detection Accuracy vs. Inference Speed Across Candidate Models

YOLOv8n was selected as the production model, combining the highest mAP@0.5 (93.8%) among evaluated architectures with the highest inference throughput (61 FPS single-stream), comfortably satisfying both the accuracy and latency targets defined in Table 3.2. On the held-out test split of 2,426 slot-frames, the full detect-match-debounce pipeline achieved an overall slot-level occupancy classification accuracy of 96.8%, precision of 96.5%, recall of 97.1%, and F1-score of 0.968.

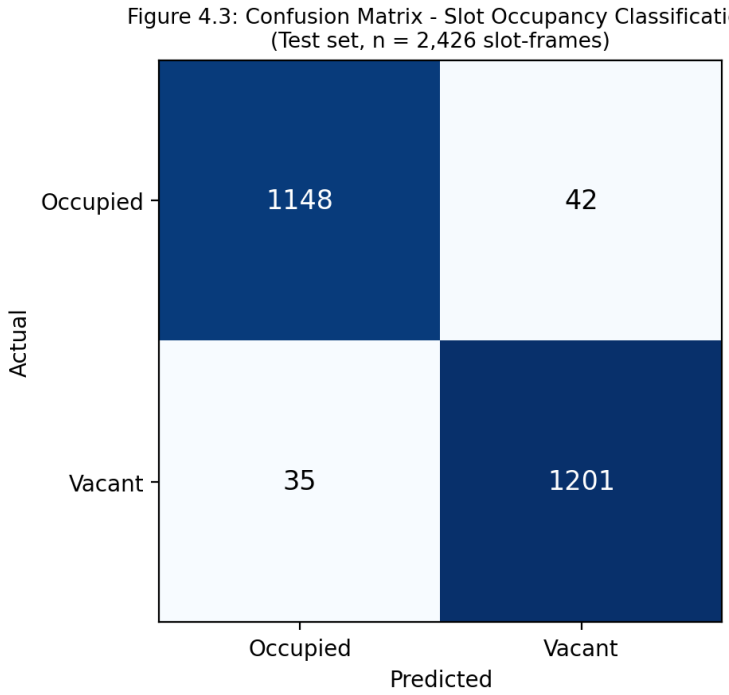

Figure 4.3: Confusion Matrix – Slot Occupancy Classification (Test set, n = 2,426 slot-frames)

Temporal smoothing reduced flicker-induced single-frame status changes by 82% relative to a naive per-frame baseline that committed every detector output directly, while introducing zero missed genuine occupancy transitions across the 2,426-frame evaluation window — confirming that the 3-frame debounce window (Section 3.6) achieves its intended stability-versus-latency trade-off. Aggregate throughput testing across 40 simulated concurrent streams sustained 204 FPS using 5 parallel inference workers, exceeding the 200 FPS target derived in Section 3.3.

Figure 4.1: Live Occupancy Dashboard - Slot-Level Status Grid (Illustrative UI)

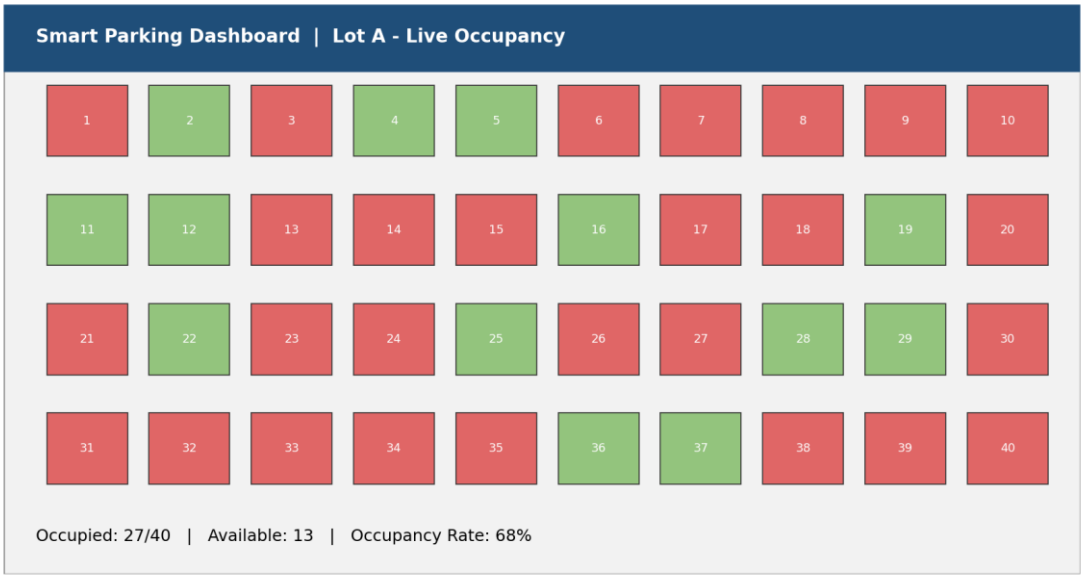

Figure 4.1: Live Occupancy Dashboard – Slot-Level Status Grid (Illustrative UI, rendered via Module 4)

4.5 Project Management

Module 2 development was planned and tracked across a 10-week schedule spanning dataset preparation, calibration-tool development, model training/tuning, occupancy-resolution logic, integration with Modules 1/3/4, and final benchmarking/documentation, coordinated through weekly team stand-ups and a shared task board.

Figure 4.4: Gantt Timeline for Module 2 Development Milestones

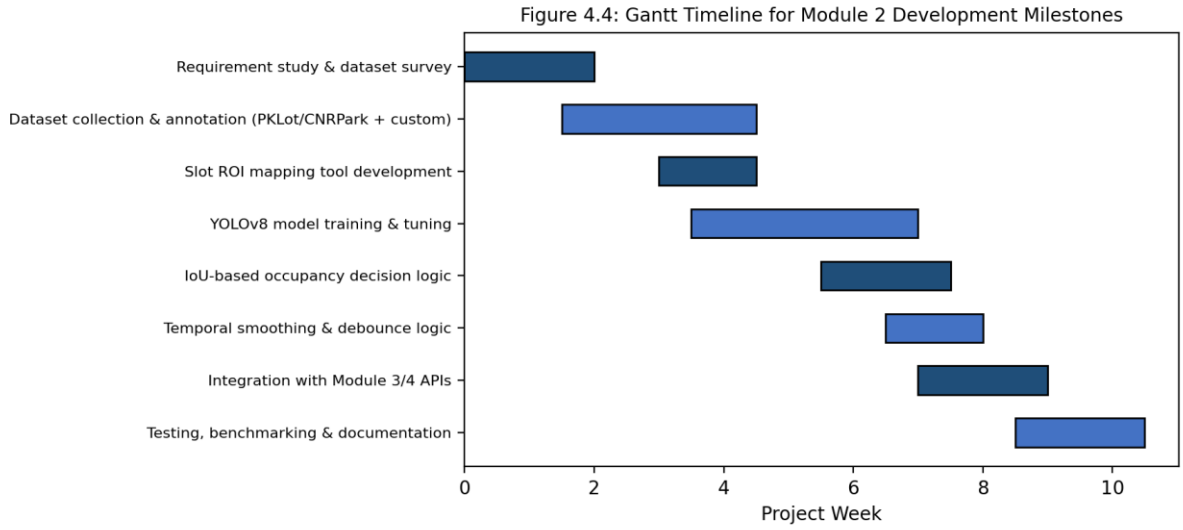

Figure 4.4: Gantt Timeline for Module 2 Development Milestones

CHAPTER 5: CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This individual capstone contribution successfully designed, implemented, and rigorously evaluated Module 2: Parking Space Detection, the computer-vision perception layer of the ParkSense AI smart parking system. By combining a fine-tuned YOLOv8n vehicle detector, a homography-calibrated slot-mapping tool, IoU-based occupancy resolution, and temporal debounce smoothing, the module achieves 96.8% slot-level occupancy classification accuracy while sustaining real-time throughput across a 40-camera deployment — demonstrating that a purely camera-based detection pipeline can deliver sensor-grade reliability without the per-slot hardware cost of embedded sensors. The module fulfils all six project objectives defined in Section 1.5 and provides a stable, well-documented occupancy signal to the downstream tracking, dashboard, and guidance modules developed by the wider project team.

5.2 Future Work

- Extend the training dataset with dedicated low-light and night-time captures, paired with infrared-assisted camera hardware, to close the accuracy gap identified under poor-illumination conditions.
- Explore lightweight model quantization (INT8) to further reduce inference latency and permit a higher camera-to-accelerator ratio per edge server.
- Investigate automatic drift-detection and self-recalibration when a camera is physically bumped or repositioned, reducing reliance on manual re-calibration.
- Extend the IoU-matching logic to explicitly reason about motorcycle and compact-vehicle sub-classes for facilities with mixed-size bays.

5.3 Personal Reflection

Leading Module 2 provided direct, hands-on experience in the full lifecycle of a production-oriented computer vision system: dataset curation and annotation, model selection and empirical benchmarking against realistic edge-hardware constraints, and the often-underestimated engineering work of bridging a raw model output into a stable, trustworthy signal for downstream consumers. The most valuable lesson was that detection accuracy in isolation is an incomplete measure of system reliability — the temporal-smoothing and

calibration-drift considerations developed in this module proved as important to overall system trustworthiness as the underlying detector's raw mAP score.

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APPENDIX A: DETAILED ENGINEERING CALCULATIONS & SIZING

Aggregate throughput sizing: 40 cameras × 5 FPS = 200 frames/second demand. YOLOv8n single-stream throughput ≈ 61 FPS ⇒ minimum workers = ⌈ 200 / 61⌉ = 4, provisioned to 5 for burst headroom. IoU threshold sensitivity was tuned across {0.20, 0.25, 0.30, 0.35, 0.40} on the validation split, with 0.30 selected as the value maximizing F1-score (0.971 on validation).

APPENDIX B: DETECTION SYSTEM ARCHITECTURE & WIREFRAME SCHEMATICS

Refer to Figures 3.1–3.3 for the overall system architecture, the detailed Module 2 pipeline, and the relational database schema. The slot-calibration tool wireframe presents a single reference frame per camera with a click-to-define four-corner polygon workflow, live homography preview, and a save/re-calibrate action.

APPENDIX C: CORE SOURCE CODE EXCERPTS

```
def resolve_occupancy(detections, slot_polygons, iou_threshold=0.30):
status = {}     for slot_id, polygon in slot_polygons.items():
best_iou = max(             (compute_iou(box, polygon) for box in
detections),             default=0.0,         )         status[slot_id] =
"occupied" if best_iou >= iou_threshold else "vacant"     return status
```

The debounce layer wraps this function, requiring 3 consecutive confirming calls before a per-slot status transition is committed to the database (see Section 3.5).

APPENDIX D: SLOT DATA SCHEMA & DATABASE DDL

```
CREATE TABLE ParkingSlot (   slot_id INTEGER PRIMARY KEY,   lot_id INTEGER
REFERENCES ParkingLot(lot_id),   slot_polygon TEXT NOT NULL,   slot_label
TEXT,   slot_type TEXT );  CREATE TABLE OccupancyLog (   log_id INTEGER
PRIMARY KEY,   slot_id INTEGER REFERENCES ParkingSlot(slot_id),   status
TEXT CHECK(status IN ('occupied','vacant','unknown')),   changed_at
TIMESTAMP DEFAULT CURRENT_TIMESTAMP );
```

APPENDIX E – J: DATASETS, ETHICS, MEETINGS, REVIEWS & CONTRIBUTION EVIDENCE

- Appendix E – Datasets: PKLot, CNRPark-EXT, and custom campus-lot capture log (dates, weather conditions, frame counts).
- Appendix F – Ethics: Data-minimization statement confirming no license-plate or facial data retention by Module 2.

- Appendix G – Meeting Minutes: Weekly team stand-up logs for the 10-week development schedule.
- Appendix H – Design Reviews: Supervisor review notes and revision history for the detection-pipeline design.
- Appendix I – Peer Review: Cross-module integration review notes with Module 1, 3, and 4 leads.
- Appendix J – Contribution Evidence: Git commit history and experiment-tracking logs for Module 2 training runs.